

Yuhao Cheng

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Education

University of Illinois Urbana-Champaign

Master of Computer Science, GPA 3.85/4.00

BS in Computer Engineering, GPA 3.77/4.00

Champaign, IL

Aug 2025 – Dec 2026

Aug 2021 – May 2025

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C, C++

Backend & Infrastructure: FastAPI, Docker, Linux, Git, PostgreSQL, Redis, Prometheus, Locust, AWS, GCP, MySQL, Nginx

Frontend: Vue 3, ECharts, Vitest, React, Node.js, Express

ML Systems: PyTorch, DeepSpeed, Accelerate, SLURM, Weights & Biases

Industry Experience

visibilityx.ai

Jun 2025 – Aug 2025

Frontend Developer Intern

- Developed a **Vue 3 and TypeScript** single-page application with 10+ data-intensive dashboard views; extracted 12 shared components and composables, removing about 2,000 lines of duplicated view logic.
- Cut median dashboard load time from 2.4 s to 1.6 s by issuing REST calls in parallel, caching responses in the Pinia store, and lazy-loading ECharts modules per route.
- Wrote **Vitest** unit tests for core frontend modules (180+ tests, 80% line coverage) and ran them in CI on every pull request to catch regressions.

HiABR Lab

May 2024 – Aug 2024

Backend Developer Intern

- Deployed a self-hosted text-to-image model (Stable Diffusion XL) as a **multi-node, multi-GPU** inference service (2 nodes, 8 GPUs, one FastAPI worker per GPU) behind a **load balancer** (Nginx, least-connections routing with health checks).
- Traced GPU out-of-memory crashes to overlapping requests running inference on the same 24 GB GPU; fixed them by routing each request through a bounded per-GPU job queue with a single worker, then added dynamic micro-batching (up to 4 prompts per 50 ms window) to win back the queueing delay.
- Load-tested with Locust: the service went from failing above 4 concurrent users to 64 with zero errors, and throughput scaled from 6 images/min on one GPU to 140 images/min across 8 GPUs at 21 s p95 time-to-image.
- Built a URL-shortening service with FastAPI, PostgreSQL, and Redis, with sliding-window rate limiting, idempotency keys, Docker, and Prometheus metrics.

Research Engineering Experience

TIMAN Group, University of Illinois Urbana-Champaign

May 2026 – Sep 2026

Research Assistant

- Built a Python evaluation pipeline that executed **3M+ LLM calls** across 8 models on a **multi-threaded** worker pool (ThreadPoolExecutor, 64 workers) with a token-bucket rate limiter and exponential-backoff retries.
- Load-tested the pipeline to find the highest concurrency the model endpoints sustained without rate-limit errors, raising throughput from 40 to 2,400 calls/min and cutting a full run from an estimated 52 days to 21 hours.
- Made runs resumable with request-level caching and checkpointing, so a failed job restarts without re-issuing completed calls; stored all generations, scores, and metrics in **Weights & Biases** for reproducibility.

IBM-Illinois Discovery Accelerator Institute

Jul 2024 – Dec 2024

Research Intern

- Scaled **full-parameter training** of Llama-3.1-8B across multi-node, multi-GPU **SLURM** clusters (4 nodes, 16 GPUs) using PyTorch, Accelerate, and DeepSpeed ZeRO-3.
- Built an SFT-to-RL training pipeline integrating browser interaction, trajectory collection, and automated evaluation across 800+ WebArena tasks; parallelized rollouts across 16 browser workers, cutting a full evaluation pass from 20 h to 3 h.

Projects and Publications

PIR-Arena: Proactive Information Recommendation Benchmark

May 2026 – Present

- Built a retrieval pipeline connecting LLM-based query generation with **BM25, dense retrieval, and cross-encoder reranking** over a 50,000-document knowledge base, at 1.8 s p95 end-to-end latency.

VGI-Bench: Probing Visual Intelligence in Video Generation Models

EMNLP 2026

- Built pipelines automating inference, output collection, and analysis for **9 video and 11 image generation models** across a benchmark of **27 tasks and 810 instances**.